CKD Lab Features Dictionary (60 Features)

1. Kidney Function (5 features)

Key	Name	Description	Min	Max	Unit	Conversion
Serum_Creatinine	Serum Creatinine	Waste product from muscle metabolism; primary marker for kidney function	0.2	20.0	mg/dL	1 mg/dL = 88.4 µmol/L; 1 µmol/L = 0.0113 mg/dL
BUN	Blood Urea Nitrogen	Measures urea nitrogen in blood; reflects protein metabolism and kidney function	5.0	200.0	mg/dL	1 mg/dL BUN = 0.357 mmol/L urea; 1 mmol/L urea = 2.8 mg/dL BUN
eGFR	Estimated Glomerular Filtration Rate	Calculated estimate of kidney filtration capacity; defines CKD staging	0.0	150.0	mL/min/1.73m²	Same in SI units
Cystatin_C	Cystatin C	Alternative GFR marker; less affected by muscle mass than creatinine	0.4	10.0	mg/L	Same in SI units (mg/L)
BUN_Creatinine_Ratio	BUN to Creatinine Ratio	Helps differentiate prerenal, renal, and postrenal azotemia	5.0	50.0	ratio	Dimensionless ratio; normal 10-20:1

2. Electrolytes (6 features)

Key	Name	Description	Min	Max	Unit	Conversion
Sodium	Sodium	Major extracellular cation; regulates fluid balance and blood pressure	110.0	170.0	mEq/L	1 mEq/L = 1 mmol/L (monovalent ion)
Potassium	Potassium	Major intracellular cation; critical for cardiac and muscle function; often elevated in CKD	2.0	8.5	mEq/L	1 mEq/L = 1 mmol/L (monovalent ion)
Chloride	Chloride	Major extracellular anion; important for acid-base balance	70.0	140.0	mEq/L	1 mEq/L = 1 mmol/L (monovalent ion)
Bicarbonate	Bicarbonate (CO2)	Primary blood buffer; often low in CKD due to metabolic acidosis	5.0	40.0	mEq/L	1 mEq/L = 1 mmol/L
Anion_Gap	Anion Gap	Calculated value (Na - Cl - HCO3); helps classify metabolic acidosis	0.0	40.0	mEq/L	1 mEq/L = 1 mmol/L
Magnesium	Magnesium	Essential mineral for enzyme function; levels vary in CKD	0.5	10.0	mg/dL	1 mg/dL = 0.411 mmol/L; 1 mmol/L = 2.43 mg/dL; 1 mg/dL = 0.823 mEq/L

3. Mineral & Bone Metabolism (7 features)

Key	Name	Description	Min	Max	Unit	Conversion

Serum_Calcium	Serum Calcium (Total)	Total calcium including protein-bound; important for CKD-MBD managem ent	5.0	15.0	mg/dL	1 mg/dL = 0.25 mmol/L; 1 mmol/L = 4.0 mg/dL; 1 mg/dL = 0.5 mEq/L
Ionized_Calcium	Ionized Calcium	Physiologically active calcium fraction; more accurate in hypoalbuminemi a	0.8	2.0	mmol/L	1 mmol/L = 4.0 mg/dL; 1 mg/dL = 0.25 mmol/L
Phosphorus	Phosphorus (Inorganic Phosphate)	Often elevated in CKD; key target in CKD-MBD management	1. 0	15.0	mg/dL	1 mg/dL = 0.323 mmol/L; 1 mmol/L = 3.1 mg/dL
Intact_PTH	Intact Parathyroid Hormone	Regulates calcium/phosphorus; often elevated in CKD (secondary hyp erparathyroidism)	5.0	2500.0	pg/mL	1 pg/mL = 0.106 pmol/L; 1 pmol/L = 9.43 pg/mL
Vitamin_D_25OH	25-Hydroxyvitamin D	Storage form of vitamin D; often deficient in CKD	4.0	200.0	ng/ mL	1 ng/mL = 2.496 nmol/L; 1 nmol/L = 0.4 ng/mL
Vitamin_D_1_25OH	1,25-Dihydroxyvitamin D (Calcitriol)	Active form of vitamin D; produced by kidneys; dec reased in CKD	5.0	100.0	pg/mL	1 pg/mL = 2.6 pmol/L; 1 pmol/L = 0.385 pg/mL
Alkaline_Phosphatase	Alkaline Phosphatase	Enzyme from bone/liver; marker of bone turnover in CKD-MBD	2 0.0	2000.0	U/L	Same in SI units (U/L = IU/L)

4. Hematology / CBC (5 features)

Key	Name	Description	Min	Max	Unit	Conversion
Hemoglobin	Hemoglobin	Oxygen-carrying protein; often low in CKD due to reduced erythropoietin	3.0	20 .0	g/dL	1 g/dL = 10 g/L = 0.6206 mmol/L; 1 mmol/L = 1.611 g/dL
Hematocrit	Hematocrit	Percentage of blood volume occupied by red blood cells	10.0	65.0	%	% = L/L × 100; 1% = 0.01 L/L
RBC_Count	Red Blood Cell Count	Number of red blood cells per volume of blood	1.5	7.0	million/µL	1 million/µL = 10^12/L = 1 × 10^6/µL
WBC_Count	White Blood Cell Count	Total leukocyte count; marker of infection/inflammation	0.5	50.0	×10³/µL	1 ×10³/µL = 10^9/L = 1000/µL
Platelet_Count	Platelet Count	Number of platelets; important for bleeding risk assessment	10.0	1500. 0	×10³/µL	1 ×10³/µL = 10^9/L

5. Iron Studies (5 features)

Key	Name	Description	Min	Max	Unit	Conversion
Serum_Iron	Serum Iron	Circulating iron bound to transferrin	10.0	300.0	µg/dL	1 µg/dL = 0.179 µmo l/L; 1 µmol/L = 5.59 µg/dL
TIBC	Total Iron Binding Capacity	Indirect measure of transferrin; capacity to bind iron	50.0	600.0	µg/dL	1 µg/dL = 0.179 µmol/L; 1 µmol/L = 5.59 µg/dL
Transferrin_Saturation	Transferrin Saturation (TSAT)	Percentage of transferrin bound to iron; key for ES A therapy in CKD	0.0	100.0	%	Calculated: (Serum Iron / TIBC) × 100
Ferritin	Ferritin	Iron storage protein; target for anemia management in CKD	5.0	5000.0	ng/mL	1 n g/mL = 1 µg/L = 2.247 pmol/L
Reticulocyte_Count	Reticulocyte Count	Immature red blood cells; indicates bone marrow response	0.1	1 0.0	%	Absolute count = % × RBC count

6. Liver Function (2 features)

Key	Name	Description	Min	Max	Unit	Conversion
Total_Bilirubin	Total Bilirubin	Breakdown product of hemoglobin; marker of liver function	0.1	30.0	mg/dL	1 mg/dL = 17.1 µmol/L; 1 µmol/L = 0.058 mg/dL
Direct_Bilirubin	Direct (Conjugated) Bilirubin	Water-soluble bilirubin processed by liver	0.0	15.0	mg/dL	1 mg/dL = 17.1 µmol/L; 1 µmol/L = 0.058 mg/dL

7. Proteins / Nutrition (4 features)

Key	Name	Description	Min	Max	Unit	Conversion

Albumin_Serum	Serum Albumin	Major plasma protein; marker of nutrition and inflammation in CKD	1.0	6.5	g/dL	1 g/dL = 10 g/L; 1 g/L = 0.1 g/dL
Total_Protein	Total Protein	Sum of albumin and globulins in serum	3.0	12.0	g/dL	1 g/dL = 10 g/L; 1 g/L = 0.1 g/dL
Prealbumin	Prealbumin (Transthyretin)	Short half-life protein; sensitive marker of nutritional status	5.0	50.0	mg/dL	1 mg/dL = 10 mg/L; 1 mg/dL = 0.01 g/L
CRP	C-Reactive Protein	Acute phase reactant; marker of inflammation common in CKD	0.0	50.0	mg/L	1 mg/L = 0.1 mg/dL; 1 mg/dL = 10 mg/L

8. Lipid Panel (5 features)

Key	Name	Description	Min	Max	Unit	Conversion
Total_Cholesterol	Total Cholesterol	Sum of all cholesterol fractions; cardiovascular risk marker	50.0	500.0	mg/dL	1 mg/dL = 0.0259 mmol/L; 1 mmol/L = 38.67 mg/dL
LDL_Cholesterol	LDL Cholesterol	Low-density lipoprotein; primary target for CV risk reduction	10.0	300.0	mg/dL	1 mg/dL = 0.0259 mmol/L; 1 mmol/L = 38.67 mg/dL
HDL_Cholesterol	HDL Cholesterol	High-density lipoprotein; "good" cholesterol	10.0	150.0	mg/dL	1 mg/dL = 0.0259 mmol/L; 1 mmol/L = 38.67 mg/dL
Triglycerides	Triglycerides	Blood fat; often elevated in CKD and metabolic syndrome	30.0	2000.0	mg/dL	1 mg/dL = 0.0113 mmol/L; 1 mmol/L = 88.5 mg/dL
Non_HDL_Cholesterol	Non-HDL Cholesterol	Total cholesterol minus HDL; includes all atherogenic particles	30.0	400.0	mg/dL	1 mg/dL = 0.0259 mmol/L; 1 mmol/L = 38.67 mg/dL

9. Glucose Metabolism (3 features)

Key	Name	Description	Min	Max	Unit	Conversion
Glucose	Blood Glucose	Blood sugar level; diabetes is leading cause of CKD	20.0	800.0	mg/dL	1 mg/dL = 0.0555 mmol/L; 1 mmol/L = 18.02 mg/dL
HbA1c	Hemoglobin A1c	Glycated hemoglobin; reflects 2-3 month glucose control (may be less reliable in CKD)	3.0	20.0	%	NGSP %: eAG (mg/dL) = 28.7 × A1c - 46.7; IFCC (mmol/mol) = (NGSP% - 2.15) × 10.929
Fructosamine	Fructosamine	Glycated protein; reflects 2-3 week glucose control; useful when A1c unreliable	150.0	600.0	µmol/L	Same in SI units

10. Uric Acid (1 feature)

Key	Name	Description	Min	Max	Unit	Conversion
Uric_Acid	Uric Acid	Purine metabolism end product; often elevated in CKD; associated with gout	1.0	20.0	mg/dL	1 mg/dL = 59.48 µmol/L; 1 µmol/L = 0.0168 mg/dL

11. Urinalysis & Proteinuria (7 features)

Key	Name	Description	Min	Max	Unit	Conversion
Urine_Albumin	Urine Albumin	Albumin in urine; early marker of kidney damage	0.0	5000.0	mg/L	1 mg/L = 0.1 mg/dL; some labs use mg/24hr for timed collections
Urine_Creatinine	Urine Creatinine	Used to calculate protein/albumin ratios and assess collection adequacy	10.0	500.0	mg/dL	1 mg/dL = 88.4 µmol/L; 1 mmol/L = 0.113 mg/dL
Albumin_to_Creatinine_Ratio_Urine	Urine Albumin-to-Creatinine Ratio (UACR)	Key marker for CKD staging and progression; spot urine test	0.0	5000.0	mg/g	1 mg/g = 0.113 mg/mmol; 1 mg/mmol = 8.84 mg/g
Protein_to_Creatinine_Ratio_Urine	Urine Protein-to-Creatinine Ratio (UPCR)	Estimates 24-hour proteinuria from spot sample	0.0	20000.0	mg/g	1 mg/g ≈ 1 mg/day protein; 1 mg/g = 0.113 mg/mmol
Urine_Protein	Urine Protein (Total)	Total protein in urine including albumin and other proteins	0.0	3000.0	mg/dL	1 mg/dL = 10 mg/L; 24hr: mg/dL × volume (dL) = mg/24hr
Urine_pH	Urine pH	Acidity of urine; relevant for stone risk and acid-base status	4.0	9.0	pH units	Dimensionless; same globally

| Urine_Specific_Gravity | Urine Specific Gravity | Concentration of urine; reflects hydration and concentrati
ng ability | 1.000 | 1.040 | ratio | Dimensionless ratio; ~correlates with osmolality |

12. Cardiac Markers (4 features)

Key	Name	Description	Min	Max	Unit	Conversion
BNP	B-type Natriuretic Peptide	Cardiac hormone; elevated in heart failure and volume overload (higher ba
seline in CKD)	0.0	5000.0	pg/mL	1 pg/mL = 0.289 pmol/L; 1 pmol/L = 3.46 pg/mL
NT_proBNP	N-terminal pro-BNP	Inactive BNP fragment; more stable than BNP; elevated in CKD	0.0	35000.
0	pg/mL	1 pg/mL = 0.118 pmol/L; 1 pmol/L = 8.47 pg/mL
Troponin_I	Troponin I	Cardiac muscle protein; marker of myocardial injury (may be elevated in CKD)	0.0
50.0	ng/mL	1 ng/mL = 1 µg/L = 1000 ng/L; 1 ng/L = 0.001 ng/mL
Troponin_T	Troponin T	Cardiac muscle protein; marker of myocardial injury (often elevated in ESRD)	0.0
10.0	ng/mL	1 ng/mL = 1 µg/L = 1000 ng/L; 1 ng/L = 0.001 ng/mL

13. Thyroid Function (2 features)

Key	Name	Description	Min	Max	Unit	Conversion
Free_T4	Free Thyroxine (T4)	Unbound active thyroid hormone	0.2	6.0	ng/dL	1 ng/dL = 12.87 pmol/L;
1 pmol/L = 0.078 ng/dL
Free_T3	Free Triiodothyronine (T3)	Most active thyroid hormone; may be low in sick euthyroid syndrome
1.0 | 10.0 | pg/mL | 1 pg/mL = 1.536 pmol/L; 1 pmol/L = 0.651 pg/mL |

14. Blood Gases / Acid-Base (4 features)

Key	Name	Description	Min	Max	Unit	Conversion
pH_Arterial	Arterial Blood pH	Blood acidity; often low in CKD due to metabolic acidosis	6.8	7.8	pH
units	Dimensionless; same globally
pCO2_Arterial	Arterial pCO2	Partial pressure of carbon dioxide; respiratory component of acid-base	15.
0	100.0	mmHg	1 mmHg = 0.133 kPa; 1 kPa = 7.5 mmHg
pO2_Arterial	Arterial pO2	Partial pressure of oxygen; oxygenation status	30.0	500.0	mmHg	1 mmHg
= 0.133 kPa; 1 kPa = 7.5 mmHg
Lactate	Lactate	Anaerobic metabolism marker; elevated in hypoperfusion/sepsis	0.3	20.0	mmol/L	1
mmol/L = 9.01 mg/dL; 1 mg/dL = 0.111 mmol/L

15. Dialysis-Specific (2 features)

Key	Name	Description	Min	Max	Unit	Conversion
Beta2_Microglobulin	Beta-2 Microglobulin	Middle molecule; accumulates in dialysis; associated with amylo
idosis	0.5	50.0	mg/L	Same in SI units (mg/L)
Aluminum	Aluminum	Toxic metal; can accumulate in dialysis patients; causes bone disease	0.0	200.0
µg/L | 1 µg/L = 0.0371 µmol/L; 1 µmol/L = 26.98 µg/L |